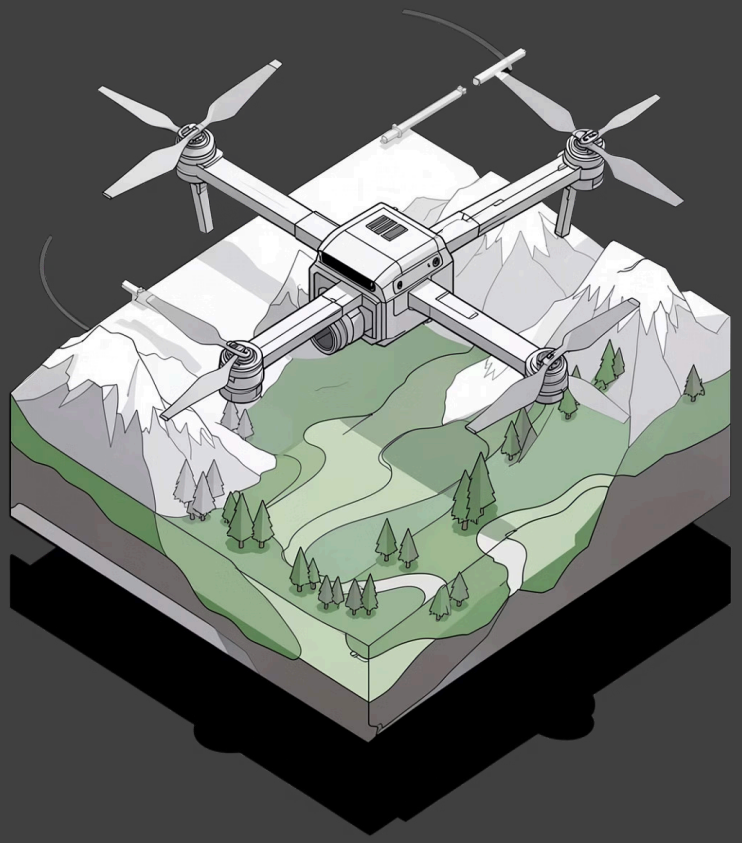

Autonomous Drone Navigation Using Image Processing

An innovative approach to enabling drones to detect obstacles, find optimal paths, and identify safe landing zones through automated image analysis.

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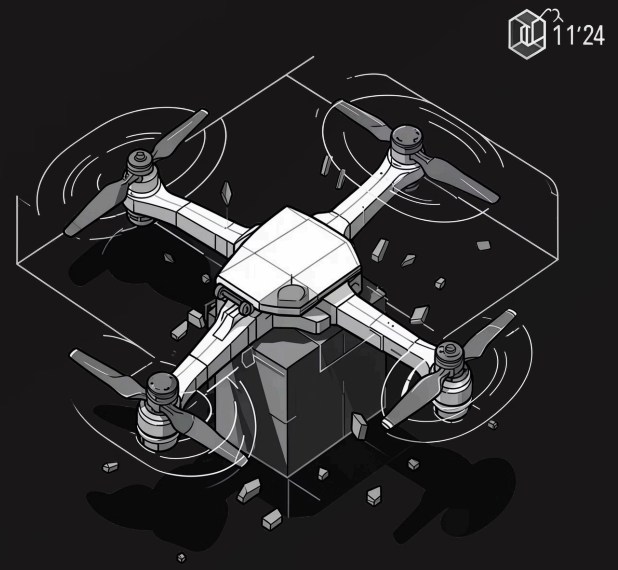
The Challenge



Drones are deployed across diverse sectors including package delivery, surveillance operations, and search-and-rescue missions. However, current systems face significant operational challenges that limit their autonomy and safety.

Key Problems

- In-flight collisions with unforeseen obstacles
- Difficulty identifying safe navigation routes
- Unsafe landing zone identification

Manual control remains unreliable for complex environments, necessitating an **automated solution**.

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Project Objectives

Obstacle Detection

Identify and mark obstacles using real-time image analysis

Path Identification

Detect navigable routes through the environment

Landing Zones

Find suitable flat areas for safe landing operations

Safety Enhancement

Improve overall drone safety through automated decision-making

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Technical Solution

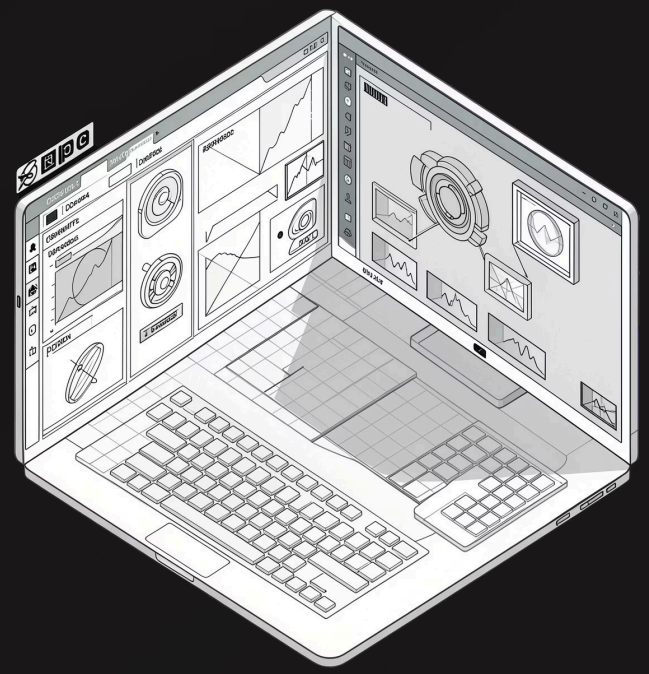

Image Processing Pipeline

The system leverages **Python and OpenCV** libraries to process visual data captured by the drone's onboard camera.

Each captured image undergoes systematic analysis to understand environmental features and make autonomous navigation decisions.

This approach eliminates the need for expensive sensors, using only standard camera hardware and computational processing.

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System Workflow

Image Capture

Camera captures environmental image

Grayscale Conversion

Reduces data complexity

Noise Reduction

Applies blur filter to smooth image

Feature Detection

Identifies edges, paths, and regions

Each processing stage progressively refines the image data, improving detection accuracy and reliability for autonomous navigation decisions.

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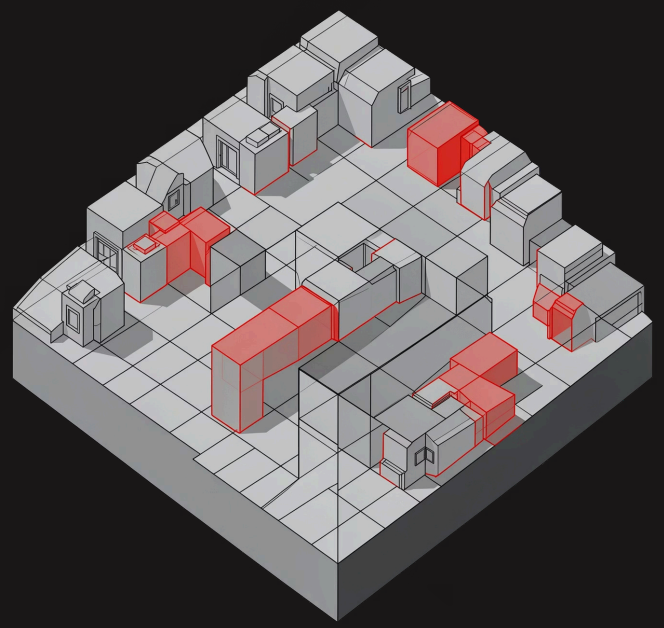
Obstacle Detection

Canny Edge Detection Algorithm

The system employs the **Canny Edge Detection** technique to identify boundaries and edges within the image. These detected edges represent physical obstacles such as walls, trees, or buildings.

The algorithm uses multi-stage processing including noise reduction, gradient calculation, and edge tracking to produce precise obstacle boundaries.

Detected obstacles are highlighted in **red colour** for clear visual identification.



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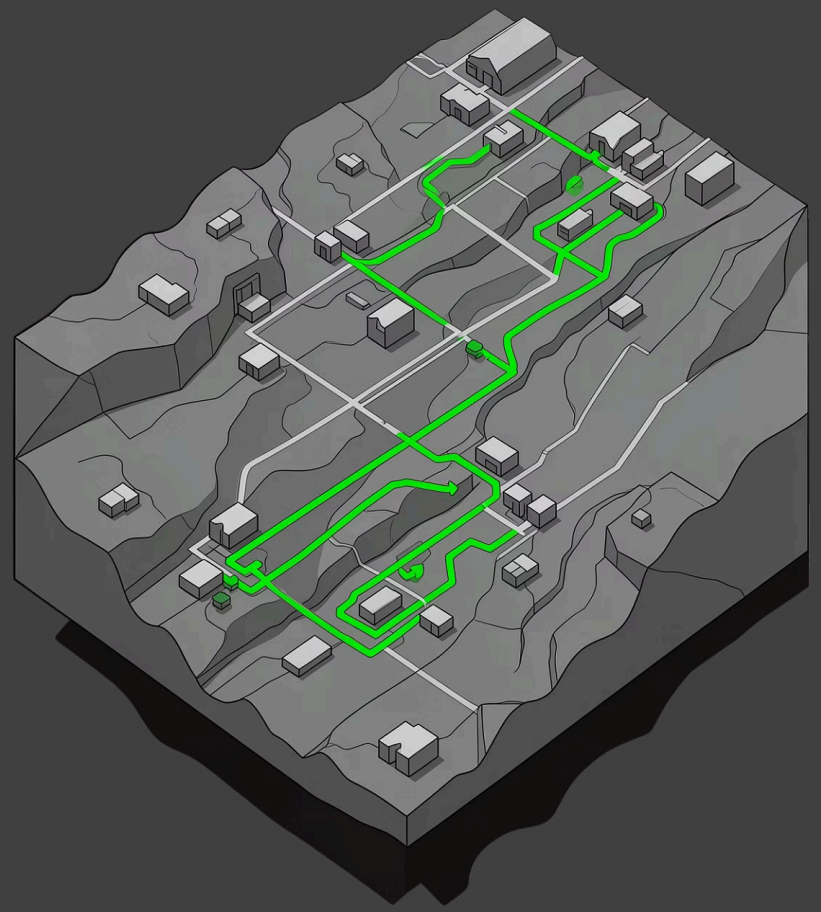

Path Detection

Thresholding Process

Converts image to binary format, identifying bright areas that represent navigable free space

Contour Detection

Analyses connected regions to determine continuous paths through the environment

Visual Highlighting

Valid navigation paths are marked with **green colour** for immediate recognition

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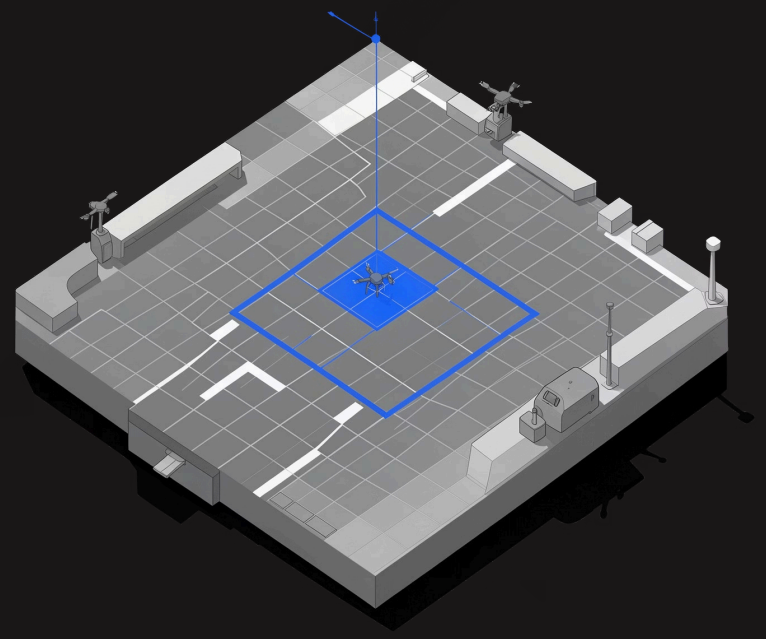
Landing Zone Detection

Identifying Safe Landing Areas

The system analyses detected contours to find the **largest contiguous flat area** within the environment. This area is considered the optimal landing zone due to its size and flat characteristics.

The detection algorithm evaluates contour shapes and sizes, selecting the region that provides maximum safety for landing operations.

The identified landing zone is marked with a **blue rectangle** for clear visual identification.



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Results & Advantages

Visual Output

Obstacles

Marked in **red**

Paths

Highlighted in **green**

Landing Zone

Outlined in **blue**

Key Advantages

- **Simple and Fast**

  Efficient processing suitable for embedded systems

- **Low Cost**

  Uses standard camera hardware without expensive sensors

- **Easy to Implement**

  Python-based implementation with OpenCV library

- **Extensible**

  Can be extended to real-time video processing systems

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Conclusion & Future Directions

Key Achievements

This project demonstrates how image processing enables drones to navigate safely, avoid obstacles, and land properly through automated decision-making.

Future Enhancements

AI Integration

Implement machine learning for improved object recognition

Real-Time Processing

Extend to continuous video stream analysis

Sensor Fusion

Combine with depth sensors for 3D mapping

Thank you for your attention.

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